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To cite this article: Shuai Zhou, Jun Zhou, Lu Li, Shicheng Zhao, Yaoqi Shi & Zhong Xin (2018): Relationship between Peroxide Initiators and Properties of Styrene Grafted Polypropylene via Reactive Extrusion, Journal of Macromolecular Science, Part B, DOI: 10.1080/00222348.2018.1467714

To link to this article: <https://doi.org/10.1080/00222348.2018.1467714>



Accepted author version posted online: 30 Apr 2018.



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Publisher: Taylor & Francis

Journal: *Journal of Macromolecular Science, Part B*

DOI: <https://doi.org/10.1080/00222348.2018.1467714>

Relationship between Peroxide Initiators and Properties of Styrene Grafted Polypropylene via Reactive Extrusion

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Relationship between initiators and PP grafting

Abstract: To understand the relationship between the initiators and the properties of grafted polypropylene (PP), and provide guidance for designing polymers with different performance through selecting appropriate initiators, a series of styrene (St) grafted PP was prepared by modifying commercial linear PP via reactive extrusion using two different peroxide initiators, dicumyl peroxide (DCP) and benzoyl peroxide (BPO). Fourier transform infrared spectra indicated that the use of DCP led to a higher St grafting degree compared to the system using BPO. The melt flow index and rheological characteristics suggested the existence of short chain branching (SCB) structures in the St grafted PP using DCP, and long chain branching (LCB) structures in the St grafted PP using BPO. Differential scanning calorimetry and polarized optical microscopy results showed that the degradation of the PP chains and the introduction of SCB structures hindered the crystallization process of the St grafted PP using DCP, and the existence of the LCB structures accelerated the crystallization process of the St grafted PP using BPO. We suggest this research can contribute to the understanding of methods to prepare grafted PP with special properties via reactive extrusion by using proper initiators.

Keywords: Polypropylene; initiators; grafting; rheological properties; crystallization behavior.

Introduction

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Polypropylene (PP) has gained significant growth in recent years due to its outstanding functional characteristics, such as excellent chemical resistance, high stiffness, high melting point, low density and low cost. However, PP also suffers some disadvantages that limit its application. For example, the nonpolar nature of PP restricts its surface wettability and interaction with polar products [1]. Moreover, the linear PP, with poor melt strength, suffers from poor processability in cases like thermoforming, film blowing, blow molding, foaming, etc., where melt stretching or drawing is usually involved [2]. Introducing short chain branching (SCB) or long chain branching (LCB) structures onto the PP backbones through reactive extrusion in the presence of appropriate polyfunctional monomers and peroxide initiators is one of the most attractive options to overcome such disadvantages because of it being a continuous and relatively easy process [1, 3-8].

In spite of the fact that reactive extrusion is an easy operation, the occurrence of two main less desirable reactions, i.e., chain scission and crosslinking, can decrease the molecular weight and processing flowability, respectively, and ultimately affect the performance of the products [9, 10]. It has been reported that these side reactions can be restrained by using an appropriate grafting monomer and initiator chemical structure [11, 12]. An electron-donating comonomer, like thiuram disulfides [13], furan derivatives [14], styrene (St) [15, 16], etc. can reduce the PP chain scission and improve the degree of grafting for PP [11, 12]; for this purpose St is the most commonly used. It has been reported that St, in the presence of free radicals, can produce more stable macroradical sites, which increase the possibility of further reactions because of a decrease of the probability of fragmentation [15, 16].

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Besides the comonomer, the initiator also plays an important role in the process of reactive extrusion because different types of initiators have different influences on the grafting polymerization [1, 17, 18]. In the grafting process organic peroxides are usually regarded as suitable initiators [18, 19]. The ROOR bonds of a peroxide undergo homolytic cleavage and release RO· radicals that may function as primary radical sources initiating these grafting reactions. Dicumyl peroxide (DCP) and benzoyl peroxide (BPO) are the most common initiators used in the PP melt grafting reaction [20-22]. Yin et al. [20] reported that DCP possessed higher initiation efficiency compared with BPO in the graft polymerization of polypropylene-graft-cardanol (CAPP), and the shear stress and viscosity of CAPP using DCP as the initiator was lower than that of CAPP by using BPO. Su et al. [21] successfully prepared long chain branching polypropylene by reactive extrusion in the presence of various peroxides and a polyfunctional monomer of 1,6-hexanediol diacrylate, and they found that the branching efficiency of BPO was higher than that of DCP. Additionally, compared to using one initiator of DCP or BPO only, the binary initiators (DCP and BPO) were proven to improve the grafting level of silane grafting of high-density polyethylene and diminish the amount of gel from the crosslinking side reaction [22]. However, the effect of these two kinds of initiators on the structure and properties of the grafted PP products has not been clear.

In the present study we focused on a fundamental study of the preparation of St grafted PP via reactive extrusion using two different peroxide initiators, DCP and BPO. The effects of initiator type on the grafting degree of St, the chain structures, the rheological properties and the crystallization behavior of St grafted PP were systematically investigated. Furthermore, we sought to find the mechanism of the grafting reaction with the two kinds of initiators. The aim of this work was to obtain a comprehensive understanding of the

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relationship between the initiator type and the properties of grafted PP. In addition, we suggest this study could also provide guidance for designing polymers with different performances through selecting appropriate initiators.

Experimental

Materials

iPP powder (T30S) was supplied by Jiujiang Petrochemical Corp., China; it had a melt flow index of 3.2 g/10min (230°C, 2.16 Kg) and density of 0.9 g/cm³. Styrene (St), dicumyl peroxide (DCP), benzoyl peroxide (BPO) and the other chemical reagents were purchased from Shanghai Lingfeng Chemical Regent Co., Ltd., China. All raw materials were used as-received.

Samples preparation

Modification of iPP was conducted in a twin-screw extruder (HT32, Nanjing Rubber and Plastics Machinery Plant Co., Ltd., China). The mixtures of iPP, St and initiators were fed into the twin-screw extruder with a screw rotating speed of 120 rpm and the feed rotating speed of 20 rpm. The processing temperatures from the first zone to the ninth zone were set as 160, 180, 190, 200, 200, 200, 200, 210, and 210°C, respectively. Table 1 shows the formulation and the labels of the samples.

Characterization

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To remove unreacted St monomer and homopolymerized St (PS), 1 g modified iPP was dissolved in 100 mL boiling xylene for 30 min, and then the hot solution was immediately poured into 400 mL petroleum ether. The iPP and St grafted iPP precipitated out and were separated from the PS and any remaining St monomer by filtration and dried at 80°C under vacuum for 24 h [23]. The purified samples were analyzed by Fourier transform infrared spectroscopy to measure the grafting degree of St (FTIR, Nicolet iS10, Thermo Fisher Scientific Inc., USA).

The melt flow index (MFI) of all samples were measured using a melt flow indexer (MI-4, Gottfert Corp., Germany) at 230°C with a load of 2.16 Kg according to the ASTM D1238.

The gel contents of all modified iPP samples were measured according to ASTM D2765. The sample pellets were wrapped in a copper net of 200 mesh and extracted with refluxing xylene for 24 h; the copper net was then dried to a constant weight in a vacuum oven at 60°C. The gel content is expressed as the ratio of the weight of sample after and before refluxing. The gel determination results showed that no gel was observed in any of the modified iPP samples.

The dynamic rheology of all samples was examined on a rotational rheometer (MCR101, Anton Paar Co., Austria) with a parallel-plate fixture (25 mm diameter and 1 mm gap height) at 200°C. An oscillatory shear mode was chosen at a strain of 1%, which was confirmed to be in the linear viscoelastic region, for all samples over the frequency range from 0.04 to 100 rad/s.

The crystallization behaviors of all samples were characterized by differential scanning calorimetry (DSC, Diamond, Perkin-Elmer Inc., USA). The crystallization peak temperature studies were carried out in the DSC

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with the samples of 3-5 mg at a heating and cooling rate of 10°C /min under nitrogen condition from 50 to 200°C; the samples were initially held at an end temperature of 200°C for 5 min to erase the thermal and mechanical history, with the subsequent cooling scans recorded. Furthermore all samples were similarly characterized with the end temperatures of 200, 240 and 280°C with the cooling curves recorded.

The crystallization morphology studies were performed on a polarized optical microscope (POM, BX51, Olympus Corp., Japan) with a hot stage (THMS600, Linkam Scientific Instruments, Ltd., UK). All samples were placed between two glass sheets, and melted and pressed at 200°C for 5 min to eliminate the thermal and mechanical history, then the samples were rapidly cooled to 140°C and kept until the isothermal crystallization process was completed; photographs were taken during the crystallization process with the isothermal crystallization time being taken as the time at which the spherulites met.

Results and discussion

Effect of initiator type on reaction efficiency

Reaction efficiency is one of the most important indexes to evaluate an initiator. In the grafting process the peroxide compounds first decompose to two primary radicals and then the radical abstracts a hydrogen from the PP backbone to form PP macroradicals which may undergo subsequent reactions including β -scission and grafting with a monomer. It is clear that both DCP and BPO decomposed to two primary radicals during the initiation process [18]. Therefore, the grafting degree of the functional monomer can reflect the reaction efficiency of the initiator. In this group of

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experiments, the molar ratio of initiator to monomer in the feed was changed within the range of 0.05-0.30 with constant St concentration of 1.04 wt% in the feed.

To verify the success of the grafting reaction and measure the grafting degree of St, FTIR measurements were conducted. Figure 1 shows the FTIR spectra of all samples after purification. For all modified iPP samples, an additional absorption peak at 700 cm^{-1} appeared which is ascribed to the C-H bending vibration of benzene rings [24], indicating that St had been grafted onto the iPP backbone.

The absorbance ratio of the peaks at 700 cm^{-1} and 1380 cm^{-1} (A_{700}/A_{1380}) was regarded as the relative St grafting degree of the modified iPP samples using the different peroxides, in which the absorbance at 1380 cm^{-1} corresponded to the characteristic absorption of the CH_3 groups in iPP [16]. Figure 2 shows the A_{700}/A_{1380} data for all modified iPP samples. The relative St grafting degree was strongly influenced by the type of initiator; the A_{700}/A_{1380} values of the modified iPP using DCP were higher than those using BPO for all initiator loadings, indicating DCP was more effective than BPO in initiating the grafting reaction of St onto iPP backbones. Furthermore, the relative St grafting degree was also significantly affected by the dosing of the initiators. The A_{700}/A_{1380} exhibited a rapid increase with increasing of initiator concentration and achieved maximum values of 6.7% and 5.9% for using DCP and BPO, respectively, at the initiator/St feeding molar ratio for both of 0.25, and then decreased for 0.30. With elevation in the initiators concentration, the formation of free radicals due to the decomposition of the initiators increased, which led to an increase in the transfer of chains to the polymer backbone, thereby resulting in a higher grafting degree of St [25]. However, excessive initiators, on the one hand,

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produced higher primary radical concentration, which can accelerate the termination process of the radicals [26] and, on the other hand, can promote the self-polymerization of St and therefore decreased the probability of St grafting [25]. As a result, the A_{700}/A_{1380} value decreased.

The difference of the reaction efficiency of the two initiators can be explained by the half-life time ($t_{1/2}$) of the initiators and the processing temperatures. According to previous work, the $t_{1/2}$ of DCP and BPO at 200°C were 23.2 s and 0.78 s, respectively [22]. In this study the extrusion temperatures were in the range of 160-210°C and the average residence time of the mixtures in the extruder, running at a screw speed of 120 rpm, was about 70 s. Since the decomposition temperature of BPO is lower than that of DCP, the decay of BPO was faster and occurred before good mixing of the components in the barrel of the twin-screw extruder. Therefore, the number of reactive sites for monomer molecules using BPO were less than that using DCP, leading to the lower initiating efficiency of BPO than that of DCP. The results of our study agreed well with previous work [21, 27] where the systems with DCP initiator showed a higher grafting yield compared to the systems using BPO.

Effect of initiator type on MFI

Although the A_{700}/A_{1380} data can evaluate the reaction efficiency of the different initiators, it is unable to differentiate the resulting molecular structures of the modified iPP using the different initiators.

The MFI is a key parameter of a polymer that can reflect the molecular structures. Generally speaking, in a reactive extrusion, an increase of MFI suggests an increase of the chain mobility because of the fact that some chain degradation occurs, whereas the reduction in MFI values implies some grafting

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or cross-linking generation which results in the extension of some of the molecular chains [28]. Since the results of the gel determination tests showed that no gel was observed in all samples, it implied that there was no obvious cross-linking structure in all of the samples.

Figure 3 shows the MFI of all modified iPP using the two initiators. Compared to the 5.47 g/10min of virgin iPP, the MFI of the modified iPP using DCP increased markedly with the increase of dosing of DCP, indicating a decrease of molecular weight due to some chain degradation occurring. To the contrary, the MFI of the modified iPP using BPO decreased obviously with the increase of the dosing of BPO, implying an increase in the molecular weight attributed to some grafting being generated. Therefore, the MFI results indicated that DCP tended to cause chain degradation, while BPO favored chain extension.

Effect of initiator type on rheological properties

The linear viscoelastic behaviors are very sensitive to the chain structures of a polymer [13, 29]. To further distinguish the long chain branching (LCB) structure and short chain branching (SCB) structure in the modified iPP samples using the two initiators, the rheology measurements were implemented. The complex viscosity (η^*) is highly sensitive to the LCB structures. It has been reported that even a very few LCB structures can affect the degree of shear thinning compared with the linear polymer with similar molecular weight [30]. The plots of η^* vs angular frequency (ω) for all samples at 200°C are presented in Fig. 4. A Newtonian plateau in the curves for virgin iPP and modified iPP samples using DCP was seen at low ω , which was in good agreement with the behavior of typical linear polymers and suggested the formation of SCB

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structures in the modified iPP samples using DCP. Furthermore, the η^* values of the modified iPP samples using DCP were lower than that of virgin iPP in the test frequency range and the η^* values exhibited an evident decrease with the increase of DCP concentration, confirming the decrease of molecular weight which was caused by the chain scission during the reactive extrusion process. For the modified iPP samples using BPO, however, a shear-thinning behavior over the whole range of frequencies was detected, as shown in Fig. 4, indicating a larger amount of LCB structures was generated [4, 31].

Moreover, the higher the concentration of the BPO, the steeper was the shear thinning, implying that an increase in BPO increased the formation level of LCB and/or crosslinked structures in the modified iPP samples. However, the slope of the shear thinning decreased when increasing BPO was used, suggesting that the higher primary radical concentration led to more β -scission of the polymer chains. These conclusions were in agreement with the grafting efficiency and MFI results.

The curve of the storage modulus (G') plotted as a function of ω has also been proven to be a useful tool to investigate the effects of LCB structure on the rheological properties of a polymer [32]. In the terminal flow region (at low frequency), where only the longest relaxation times contribute to the viscoelastic behavior, G' for a linear polymer follow the well-known frequency dependence, $G' \propto \omega^2$ ($\ln G' \propto 2 \ln \omega$, slope = 2) [33]. Figure 5 shows G' plotted as a function of ω for all samples at 200°C. The virgin iPP exhibited typical terminal behavior, and its terminal slope of G' was 1.70, close to 2, indicating that the virgin iPP was a linear polymer. The modified iPP using DCP had similar terminal behaviors as that of the virgin iPP, and the terminal slopes of G' of DCP1, DCP2, DCP3 and DCP4 were 1.80, 1.89, 1.97 and 1.98, respectively.

Furthermore, the G' value decreased with the increase of DCP concentration because of an increase in

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the chain scission process. However, for the iPP modified using BPO, a notable terminal behavior deviation was found. The G' of the BPO modified iPPs were higher than that of virgin iPP at low ω , and with the increase of BPO concentration, the G' of modified iPP first increased, reaching the maximum when the BPO/St feeding molar ratio was 0.25; thereafter it decreased. Furthermore, the terminal slopes of G' of BPO1, BPO2, BPO3 and BPO4 were 0.90, 0.83, 0.40 and 0.99, respectively, decreased significantly compared to that of the virgin iPP and the DCP modified iPPs. These results indicated that the LCB structures were formed on the PP backbone, leading to the increased longer relaxation time [33], and the content of the LCB structures first increased and then decreased with the increase of BPO concentration.

The curve of the logarithm of G' versus the logarithm of the elastic modulus (G'') (Han plot) is another indication for formation of LCB structures [32]. The Han plot is independent of the melt temperature and weight average molecular weight for monodispersed molecular weight polymers [34]. For example, for linear PE and PE with short chain branches, the relation between G' and G'' is: $G'=0.00541 (G'')^{1.42}$; however, the Han plot of PE with LCBs will deviate from the Han plot of linear PE [32]. Figure 6 shows the Han plots for all samples at 200°C. The Han plots of the virgin iPP and modified iPP using DCP were almost linear and the data fit $\lg G' = 1.77 \lg G'' - 3.00$, implying that virgin iPP and modified iPP samples using DCP were linear polymers. For the Han plots of the modified iPP using BPO, an upshift from the linear virgin iPP curve was evident at lower G'' , indicating that LCB structures were present in these samples.

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The long relaxation process of the modified iPP samples can also be illustrated in the curve of imaginary viscosity (η'') versus dynamic viscosity (η') (Cole-Cole plot). Figure 7 illustrates the Cole-Cole plots for all samples at 200°C. The Cole-Cole plots of the virgin iPP and modified iPP using DCP exhibited an almost semicircular pattern, suggesting that they had the characteristics of linear molecular structures. The radius of the semicircle decreased with the increase of DCP due to the smaller molecular weight caused by more serious degradation reactions. Differently, the Cole-Cole plots of BPO modified iPP were not a semicircular pattern; they showed evident upturning at high viscosity, implying that these samples had longer relaxation times. Furthermore, this deviation was greater with increasing BPO quantity. These results were consistent with the study of Li et al., who attributed such phenomena to the existence of LCB structures leading to longer relaxation times [33]. However, when the amount of BPO was too much, the deviation degree decreased slightly due to the increase of degradation.

The presence of long chain branching in the polymers was also confirmed for the BPO modified iPP by the plot of the tangent of the loss angle (δ) versus ω [35]. The curves of $\tan \delta$ versus ω for all samples are shown in Fig. 8. For virgin iPP and the DCP modified iPP samples the terminal behavior was similar to that for liquid-like materials. Their $\tan \delta$ decreased rapidly with the increase of ω because of their linear chain structures. With the increase of DCP content, the $\tan \delta$ of the modified iPP increased quickly at low frequencies, indicating that the degradation was more and more serious. For the BPO modified iPP samples, the typical behavior for solid-like materials was shown. Their $\tan \delta$ were substantially lower than that of virgin iPP at low ω , being essentially zero over the entire range of ω , indicating the presence of long chain branches.

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The curve of δ versus the absolute value of the complex modulus ($|G^*|$) (vGP plot) is not only useful for evaluating polydispersity effects on the linear viscoelastic response of linear polymers, but also serves to illustrate the change of elasticity resulting from the macromolecular chain structures [36, 37]. A linear polymer is almost completely viscous, with the δ value approaching 90° at low G^* , whereas the decrease of the δ value at lower values of G^* is an indication of increasing elasticity and a tendency to form LCB structures [38, 39]. The vGP plots for all samples at 200°C are shown in Fig. 9. The vGP curves for virgin iPP and modified iPP samples using DCP overlapped to form a “universal” curve, and the δ values were close to 90° at the terminal value of G^* . It is well known that any deviations from the “universal” curve can be attributed to the influences of LCB structures or broadening of the molecular mass distribution [36, 37]. Since the vGP curves for virgin iPP and the DCP modified iPP samples had no significant changes in shape, it was concluded that there were no obvious changes in the molecular mass distribution and the linear structure of the DCP modified iPP with increasing DCP concentration. In other words, the DCP predominantly caused degradation by chain scission and produced SCB structures only when using DCP as initiator, rather than the introduction of LCB structures. However, the δ values of the BPO modified iPP were lower than that of the virgin iPP for the lower values of G^* , suggesting that the elasticity of the BPO modified iPP samples was higher than that of the virgin iPP and the DCP modified iPP samples, which we attribute to the introduction of LCB structures onto the iPP backbones. Furthermore, the δ value at lower G^* decreased with the increase of BPO concentration, implying that the elasticity of the modified iPP samples increased due to increasing LCB content. Besides, the δ values at lower G^* increased with excessive BPO (BPO4) due to the degradation that occurred.

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All of the rheology plots varied with the types of peroxides, suggesting that the elastic responses and the branching structures in the modified iPP samples were affected by the dosing and types of initiators; they confirmed the existence of SCB structures in the DCP modified iPP and LCB structures in the BPO modified iPP.

Grafting Mechanism

Summarizing the above described results, a proposed mechanism of free radical initiated St grafted PP via reactive extrusion using the two peroxide initiators is shown in Fig. 10. It is well recognized that the initiator first decomposes to form primary radicals and then abstracts tertiary hydrogens from the PP chains and styrene [Fig. 10, Reactions (1), (2) and (3)]. Subsequently, grafting reactions and β -scission of the PP macroradicals will occur simultaneously [18]. Due to the longer $t_{1/2}$ and the higher reaction efficiency of DCP, more primary radicals were produced during the melt extrusion process, which led to the generation of more PP macroradicals and St radicals. It has been reported that the styryl macroradicals are much more stable than the PP macroradicals due to the conjugation effect of the phenyl rings. Part of the PP macroradicals rapidly reacted with the majority of the St monomer to produce styryl macroradicals [Fig. 10, Reactions (5) and (6)], and the short side chains were introduced onto the PP backbone ultimately [Fig. 10, Reactions (7) and (8)]. Moreover, the other part of the PP macroradicals may undergo a β -scission reaction at high temperatures [Fig. 10, Reaction (4)]. This speculation could be proved from the previous results (Fig. 1 to Fig. 9). When using DCP as an initiator, although the St grafting degree was higher, the significant chain degradation resulted in the

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modified iPP with lower melt viscosity and higher MFI. In addition, the rheological results also showed that the SCB-PP was generated when DCP was used as the initiator.

On the other hand, for the modified iPP using BPO, the reaction process was the same to that of DCP.

But the number of the PP macroradicals was less than that using DCP since the decay of BPO is faster (shorter $t_{1/2}$) and occurred before good mixing of the components during the melt extrusion process.

Therefore, the formation of styryl macroradicals was less [Fig. 10, Reaction (5)] and long chain branch could form because of the high concentration of styrene monomer [Fig. 10, Reaction (6), (7) and (8)], and less β -scission reaction was occurred [Fig. 10, Reaction (4)]. Ultimately, the St grafting degree was lower and the chain degradation was controlled when using BPO as an initiator, contributing to the modified iPP with higher melt viscosity and lower MFI. Moreover, the rheological results also confirmed that the LCB structures were introduced onto the PP backbones.

This research contributed to confirm that the $t_{1/2}$ of the initiator is one of the most important factors that have to be taken into account when choosing a suitable initiator for the grafting process occurring during reactive extrusion. It was difficult for an initiator with a long $t_{1/2}$ at the processing temperature to initiate the reaction because of the small number of primary radicals generated during extrusion. However, an initiator with a short $t_{1/2}$ at the processing temperature would decompose prematurely, leading to a decrease of initiator efficiency [19, 22, 26, 40]. Thus, it is important to choose an initiator with the suitable $t_{1/2}$ for the processing conditions used in order to obtain the grafted polymer with ideal chain structures.

Effect of initiator type on crystallization behaviors

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The non-isothermal crystallization behaviors of all samples were studied by DSC; the DSC exothermal curves at a cooling rate of 10°C/min are presented in Fig. 11. Compared with virgin iPP, it was apparent that the crystallization peak temperature (T_c) of the DCP modified iPP first decreased with the increase of DCP loading, reaching a minimum value of 113.9°C at the initiator/St feeding molar ratio of 0.25 and then slightly increased. This result indicated that the degradation of PP chains and the introduction of SCB structures hindered the crystallization process of the DCP modified iPP. However, for the BPO modified iPP, a significant elevation in T_c was observed. As the BPO loading rose, T_c increased dramatically at first, reaching a maximum value of 129.8°C at the initiator/St feeding molar ratio of 0.25 and then slightly decreased. Furthermore, there was a small shoulder peak at lower temperatures on the cooling thermograms of the BPO-2 and BPO-4 modified iPP samples. It has previously been shown that the existence of LCB structures can increase the crystallization temperature of PP [33, 41] and the double crystallization peaks have been shown for some DSC cooling curves of LCBPP samples [8, 42, 43]. However, none of these reports explained how the LCB structures affected the crystallization behavior. In order to understand this phenomenon clearly, the crystallization behaviors of the BPO modified iPP samples were measured by DSC. The iPP and BPO modified iPP samples were heated to different end melting temperatures (200°C-280°C), and held at this temperature for 5min to eliminate the prior heating history. Then the temperature was reduced to 50°C at a cooling rate of 10°C/min. The results are shown in Fig. 12. For the BPO modified iPP samples, the peak at the higher temperature gradually became smaller with the increase of the end melting temperature and disappeared when the end melting temperature was 280°C. We suggest the

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results indicated that the double crystallization peaks on the DSC cooling curves of the BPO modified iPP samples resulted from the entanglements of the iPP with LCB structure. At a low end melting temperature (200°C - 240°C), the entanglement of LCB structures were much more stable because the LCB structures possessed longer relaxation times than the linear iPP molecules. Therefore, in the heating process, the motion of the free iPP chains could be restricted because of the entanglements of LCB structures, resulting in retention of some of the prior ordering of neighboring iPP segments promoting nucleation and an increased crystallization temperature, which was responsible for the appearance of the peak at the higher temperature. With the increase of the end melting temperature (280°C), the entanglements of the LCB structures would be totally unraveled, the ordering of the iPP chains being totally destroyed with no residual ordering accelerating the crystallization of iPP, resulting in the disappearance of the higher temperature peak. Therefore, their crystallization behaviors were consistent with iPP, and their crystallization peak temperature became the same as that of pure iPP.

The spherulite morphology of all samples was observed by POM. Figure 13 shows the POM micrographs of all samples crystallized under isothermal conditions at 140°C ; as shown, the initiator type and content significantly influenced the spherulite size and crystallization time of the modified iPP. For the DCP modified iPP, their spherulite diameters and crystallization times were much larger than that of the virgin iPP, and increased with increasing the DCP content from 0.015mol DCP to 0.025mol DCP. The results clearly indicated that the introduction of the SCB structures and the serious degradation using DCP depressed the nucleating efficiency, exhibiting an obvious anti-nucleation effect. These results are in agreement with the DSC data.

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However, the BPO modified iPP showed the opposite phenomenon. An abundance of spherulites were formed and the spherulite size and crystallization time were decreased significantly compared with that of the virgin iPP, which could be attributed to the LCB structures entanglements promote the crystal nucleation and therefore led to the increase in nucleation densities and the decrease in spherulite size. Furthermore, the spherulite size further decreased with increasing the BPO content due to the nucleation effect being enhanced with the LCB content increasing. These results were also in agreement with the DSC data.

Conclusions

It was concluded from this study that the structures and properties of St grafted PP were significantly affected by the type of initiator. FTIR measurements confirmed that both DCP and BPO can initiate the grafting of St onto PP through reactive extrusion, with DCP resulting in a higher grafting degree than BPO because of the longer $t_{1/2}$ of the DCP. The MFI and rheological characteristics suggested that DCP not only introduced SCB structures onto PP but also caused the chain scission of PP, which led to the higher flow ability and worse crystallization properties; these included a higher MFI, lower crystallization temperature, slower crystallization rate and larger spherulite size compared to that of virgin iPP. However, the case was totally different for the BPO system, for which LCB structures were introduced without obvious chain scission in the tested dosing range, resulting in St grafted PP with higher crystallization temperature and smaller spherulite size, which can benefit the mechanical properties of PP. This study thus provides guidance for designing co-polymers with different performance through selecting appropriate initiators.

Acknowledgement

The authors thank the Fundamental Research Funds for The Central Universities of China (Grants NO.

22A201514016 and 222201314051), and National Natural Science Funds of China (Grants NO. 21606084,

21476085 and 21306047) for financial support.

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Table Captions

Table 1 Formulation and labels of all samples

Figure Captions

Fig. 1 FTIR spectra of all samples after purification

Fig. 2 A_{700}/A_{1380} of all samples after purification

Fig. 3 MFI values of all modified iPP samples using the two initiators (The virgin iPP had an MFI of 5.47 g/10min)

Fig. 4 η^* versus ω of all samples at 200°C

Fig. 5 G' versus ω of all samples at 200°C

Fig. 6 Han plots for all samples at 200°C

Fig. 7 Cole-Cole plots of all samples at 200°C

Fig. 8 $\tan \delta$ versus ω of all samples at 200°C

Fig. 9 vGP plots of all samples at 200°C

Fig. 10 Mechanism of melt free radical initiated St grafted PP using the two peroxide initiators

Fig. 11 DSC exothermal curves of all samples at a cooling rate of 10°C /min after previously being heated to 200 °C for 5min

Fig. 12 DSC exothermal curves of iPP and BPO modified iPP samples when end melting temperature was change from 200°C to 280°C at a cooling rate of 10°C/min

Fig. 13 POM micrographs of all samples crystallized at 140°C

ACCEPTED MANUSCRIPT

Relationship between initiators and PP grafting

Table 1 Formulation and labels of all samples

Samples	BPO (mol)	DCP (mol)	St (mol)	PP (g)
iPP	0	0	0	1000
BPO-1	0.005	0	0.1	1000
BPO-2	0.015	0	0.1	1000
BPO-3	0.025	0	0.1	1000
BPO-4	0.030	0	0.1	1000
DCP-1	0	0.005	0.1	1000
DCP-2	0	0.015	0.1	1000
DCP-3	0	0.025	0.1	1000
DCP-4	0	0.030	0.1	1000

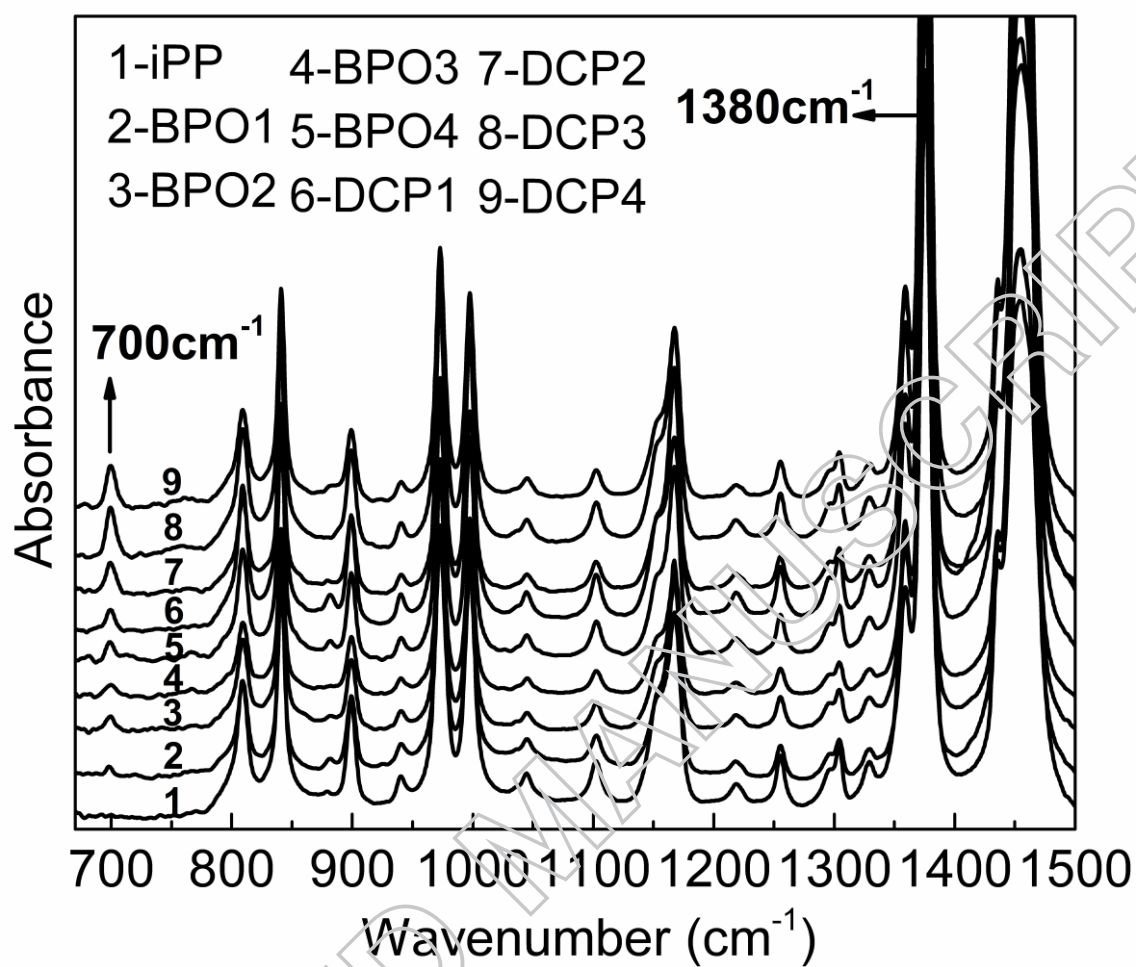


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Relationship between initiators and PP grafting

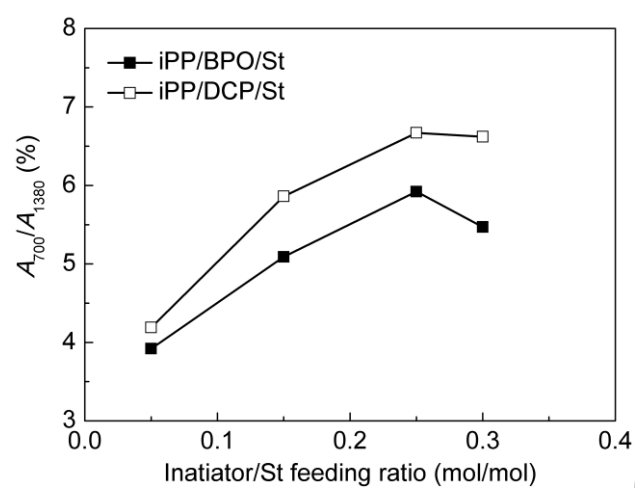


Fig. 2 A_{700}/A_{1380} of all samples after purification

Relationship between initiators and PP grafting

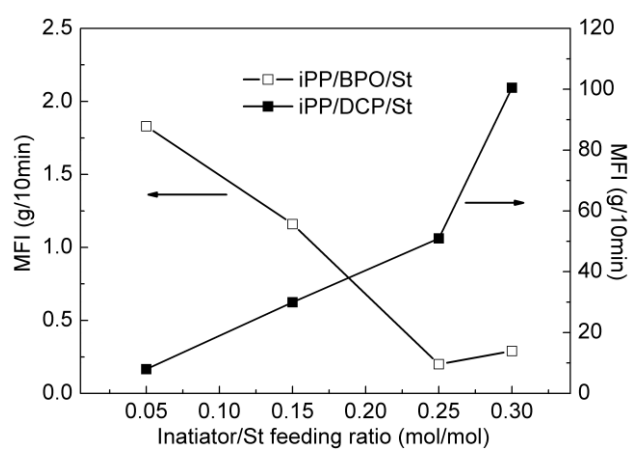


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Relationship between initiators and PP grafting

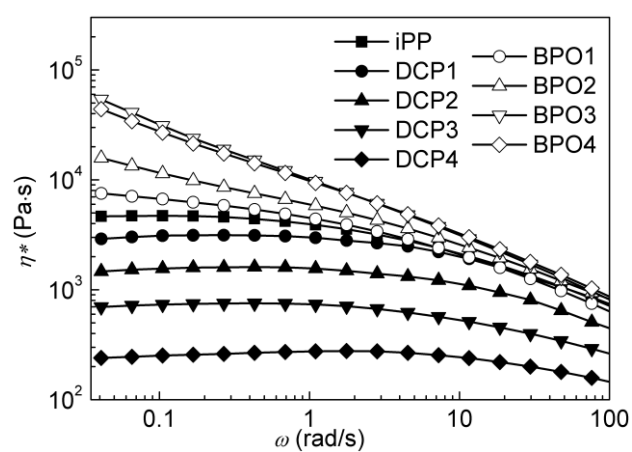
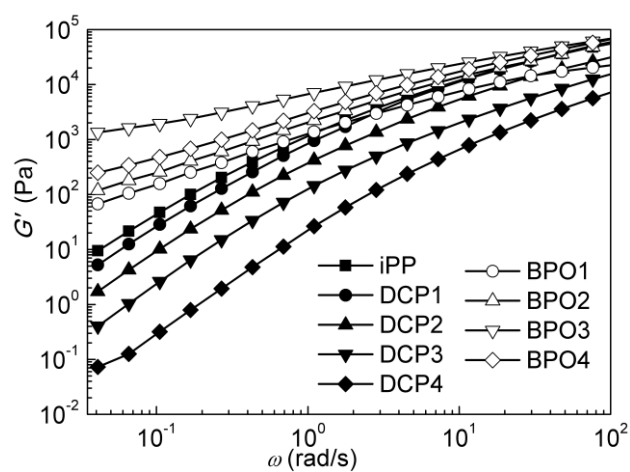


Fig. 4 η^* versus ω of all samples at 200°C

Relationship between initiators and PP grafting

Fig. 5 G' versus ω of all samples at 200°C

Relationship between initiators and PP grafting

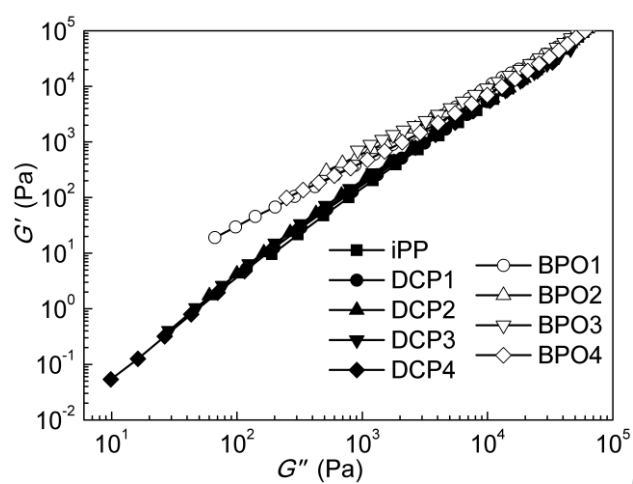


Fig. 6 Han plots for all samples at 200°C

Relationship between initiators and PP grafting

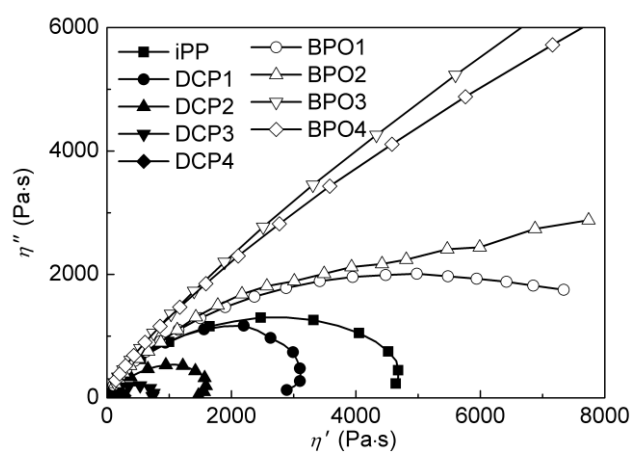


Fig. 7 Cole-Cole plots of all samples at 200°C

Relationship between initiators and PP grafting

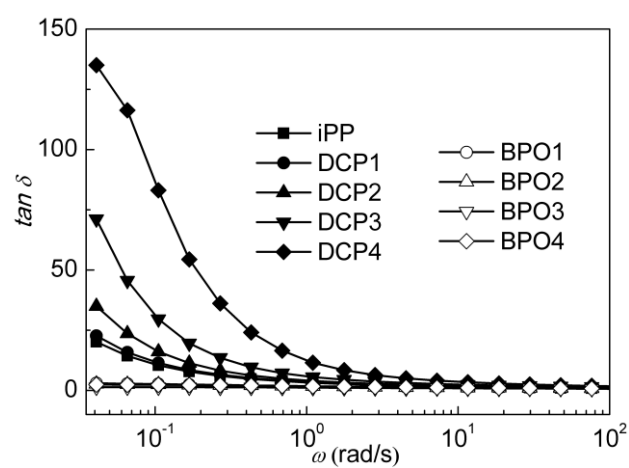


Fig. 8 $\tan \delta$ versus ω of all samples at 200°C

Relationship between initiators and PP grafting

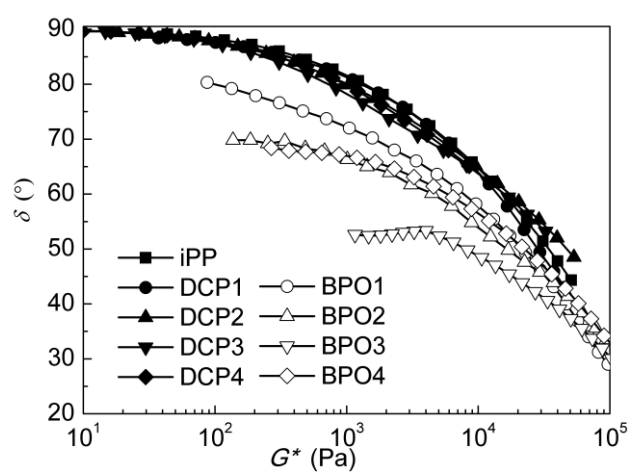


Fig. 9 vGP plots of all samples at 200°C

Relationship between initiators and PP grafting

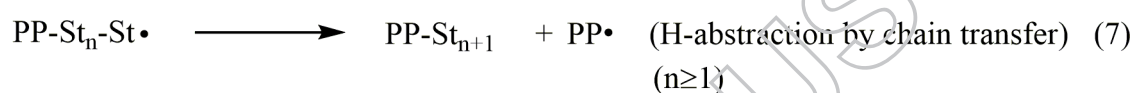
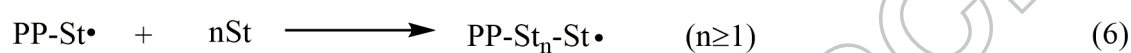
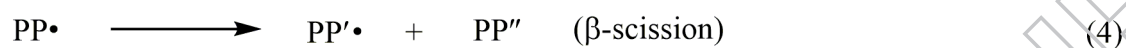


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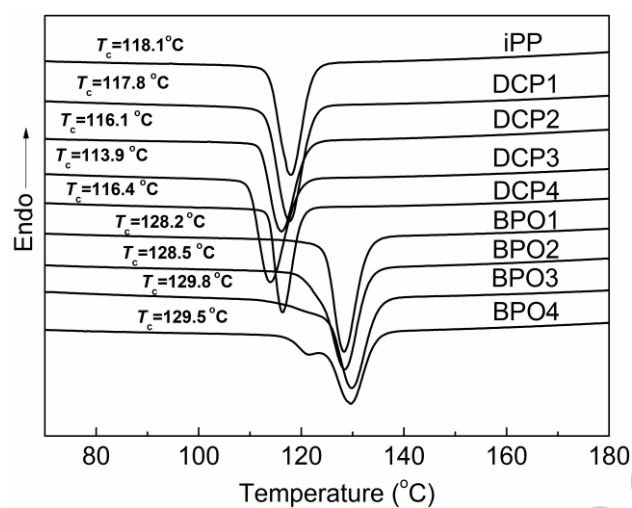


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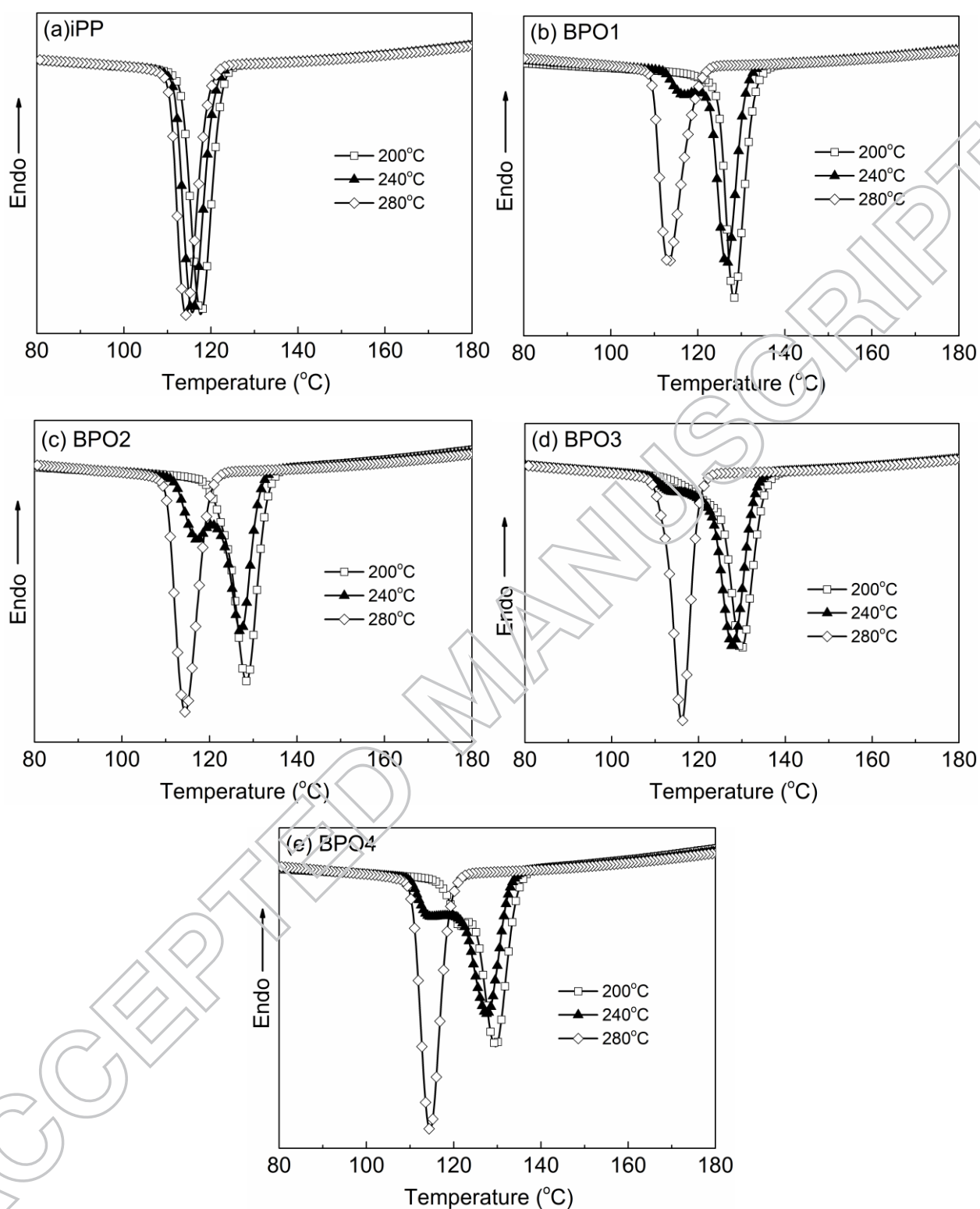


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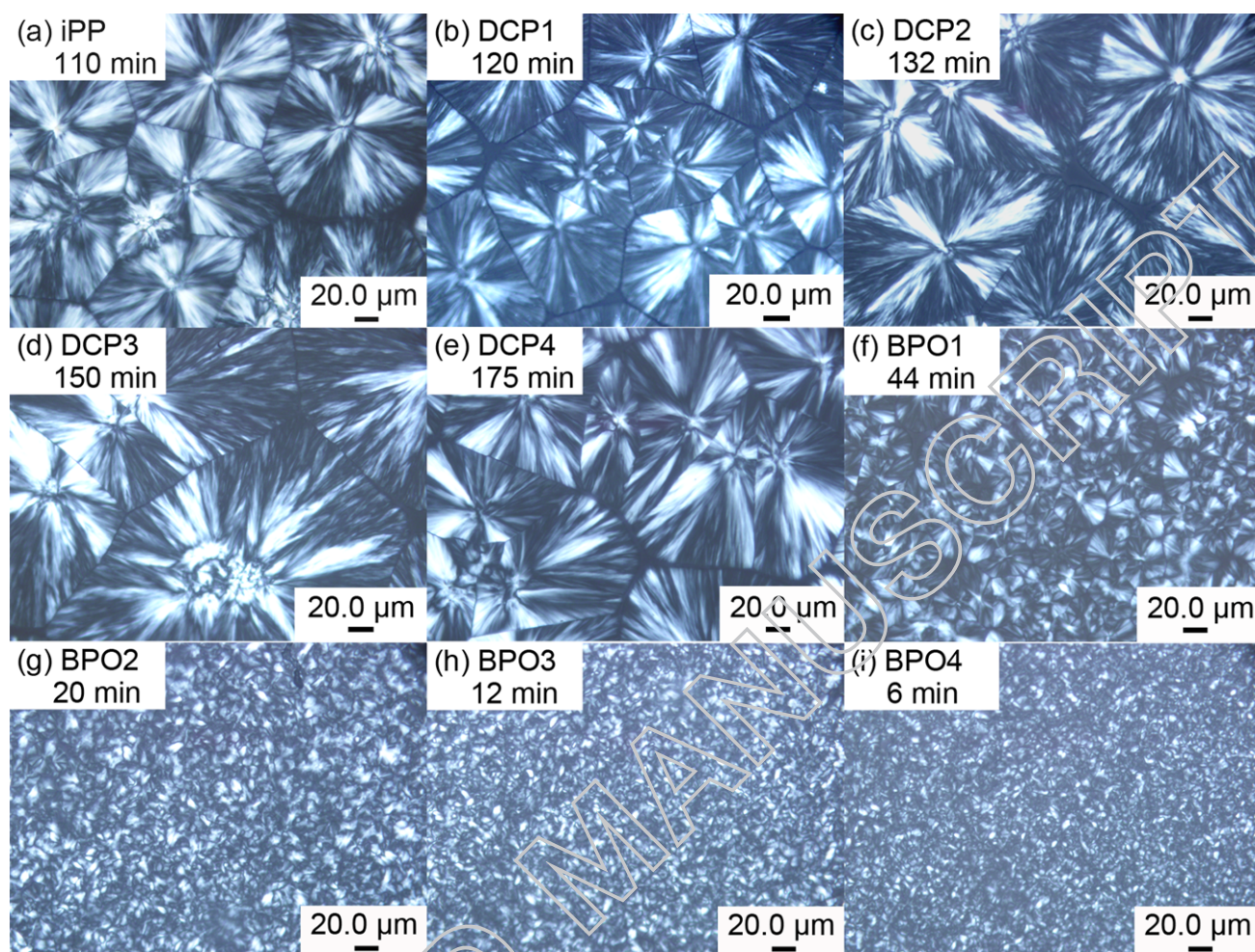


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